Harshawardhan Sachin Matkar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.84
Jr. College Mutnal	Class XII (Maharashtra state Board)	2023	86.67%
Sau. V. D. Shinde High School	Class X (Maharashtra state Board)	2021	97.80%

EXPERIENCE

Developer Intern | *Accelequant*

- **Research Implementation:** Converted academic papers on quantum error mitigation into production Python code, implementing ZNE and TREX algorithms.
- **Accuracy Optimization:** Reduced output noise in quantum simulations, achieving up to a 35% improvement in execution fidelity on noisy hardware models.
- **Quantinuum Emulation:** Built software models of Quantinuum quantum processors to simulate physical hardware limits and readout errors.
- **Code Base Upgrades:** Migrated legacy project repositories to current library versions while maintaining Git version control and writing clear documentation.

PROJECTS

Multi-Threaded HTTP Server with Custom Thread Pool | *C++17, POSIX Threads, Socket Programming, Linux*

- Built a **high-concurrency HTTP web server** in C++17 using low-level Linux TCP/IP socket APIs.
- Implemented a **custom Thread Pool** using the Producer-Consumer model with persistent worker threads to reduce thread-creation and context-switching overhead.
- Developed thread-safe task queue synchronization using **std::mutex, std::condition_variable, and std::unique_lock** to prevent race conditions.
- Implemented TCP server operations using **socket, bind, listen, and accept** to handle concurrent client connections.
- Parsed **HTTP GET requests** and served static files directly from disk.
- Applied **RAII-based resource management** to ensure automatic socket cleanup and prevent file descriptor leaks.
- Used **std::function and lambda expressions** for flexible task submission and worker execution.

Artificial Face Generator using GAN | *TensorFlow, Python, Neural Networks, CNN, GAN Architecture*

- Built a Deep Convolutional **Generative Adversarial Network** (DCGAN) in TensorFlow using standard transposed convolutions, strided convolutions, and batch normalization to synthesize 64×64 RGB anime character faces from a 100-dimensional noise vector.
- **Custom Training Loop:** Implemented standard adversarial loss using tf.GradientTape with dual Adam optimizers (= 0.0002, $_1$ = 0.5); tracked visual convergence across fixed seed vectors to monitor feature evolution over 50 epochs.
- **Quantitative Evaluation & Diagnostics:** Evaluated feature diversity by computing nearest-neighbor L2 distances in latent space against real dataset baselines (mean ratio = 1.13), confirming original image generation rather than dataset memorization.
- **Adversarial Failure Analysis:** Identified discriminator dominance (D loss 0.15 vs. G loss 4.75 by epoch 48), indicating gradient saturation; documented architectural bottlenecks (eye asymmetry, line-art bleeding) and proposed label smoothing and WGAN-GP for future stability.

RAG-based Document QA Web App | *Python, FastAPI, FAISS, Sentence-Transformers, Groq API, Streamlit*

- Built a RAG pipeline that lets users upload documents (PDF/DOCX/TXT) and ask questions about them, with answers linked back to the exact source text
- Combined FAISS vector search with BM25 keyword search for better retrieval, and added a re-ranking step to pick the most relevant chunks before sending them to the LLM
- Tested the system on a sample Q&A set and got 80% recall@5 and 0.7 MRR, showing the right chunk was usually retrieved and ranked near the top
- Built a FastAPI backend and a Streamlit chat interface so users can upload files and get answers with sources shown

alongside

SCHOLASTIC ACHIEVEMENTS

- Maharashtra State Common Entrance Test (MHT-CET) : State Rank 98
- Highest 9.58 CGPA in 3rd Year ES Department.
- Selected for Google India Launchpad Upskilling Program (2025).

SKILLS AND RELEVANT COURSES

- **Languages & Tools:** C++, Python, SQL, Git/GitHub, CMake
- **Systems Programming:** POSIX Threads, Socket Programming
- **Machine Learning:** Supervised/Unsupervised Learning, XGBoost, CNNs, RNNs, Transformers, Tensorflow/keras
- **Coursework:** Algorithms, Data Structures, OS, DBMS, Probability, Statistics, Numerical Methods, Linear Optimization

EXTRACURRICULER

Member, Canteen Monitoring Committee.

Member NSS